Correction: validation sampling

Snowball PDBThink RL report · October 21, 2026

The earlier native in-training validation was not greedy. Its configuration requested temperature 0, but the pinned MarinSkyRL serving entrypoint supplied training sampling settings to a shared wrapper. That wrapper overwrote each chat request's temperature with the training value, 1. Earlier report statements describing that curve as greedy are incorrect. This correction supersedes those statements in the historical report reproduced after this page.

Measurement	Effective sampling
Previous native training validation	Temperature 1; top-p 1; top-k disabled
Separate full step-zero baseline	Temperature 0; direct endpoint; 1,003 tasks
Matched original PDBThink benchmark	Temperature 0.6; top-p 0.95; top-k 20; direct endpoint
Corrected router-replay retry	Temperature 0; fixed 1,003-task validation from step 0

What changes: comparisons between the separate greedy baseline and native training validation also differ in sampling temperature. The paired reasoning audit remains a description of the archived responses, but is not a matched causal estimate of RL. The new greedy curve must be assessed against its own baseline; the old native curve is not a directly comparable control. This discovery does not by itself explain the late-training regression.

What remains valid: archived answers and verifier scores are unchanged. The matched original-benchmark comparison used a separately deployed endpoint and frozen requests, outside this wrapper, so those results are unaffected. The controlled, identical-token probability checks still establish that expert selection contributes to learner–rollout disagreement.

Retry changes: honor per-request sampling; request no unused probability logs for greedy validation; preserve both Snowball stop IDs explicitly; reject validation execution errors or incomplete coverage. Training uses route replay, importance weights capped at 4, fresh optimizer state, and frozen quantile biases. Its minimum-token setting is now 0 so the first-token EOS mask cannot create an unmatched sampling processor. No new RL improvement is claimed here.

The corrected step-zero evaluation scored 239/1,003 (23.83%), with no execution errors. Submission v4 then stopped because an audit guard double-counted aggregate and category totals. Submission v5 reuses that valid baseline with recorded provenance and checksum; no optimizer updates occurred before the baseline was measured.

Evidence: pinned MarinSkyRL 5663f612, skyrl_train/entrypoints/main_base.py (create_ray_wrapped_inference_engines_from_config) and inference_engines/vllm/utils.py (apply_openai_sampling); original resolved launch artifact; retry v3 validation failure and v4 request-path regression tests. See experiments/002-pdbthink-router-replay/README.md and results/baseline-step0.json for the execution record. All 277 final CPU checks passed; these are not evidence of training improvement.